OPTO

Operational manual for OpenVnmrJ

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1. Introduction:

This manual is included with the installer and can be found in the unzipped folder under the file name "`OPTO_OVJ_Manual.pdf`". The manual provides a comprehensive guide to all OPTO settings and capabilities, including an example run to demonstrate how OPTO can be utilized for OpenVnmrJ.

For the latest version of OPTO for OpenVnmrJ, go to https://git.doit.wisc.edu/nmrfam-public/opto-releases/-/blob/main/README-OVJ.md

2. Installation Guide:

a) Download the most recent `opto-ovj.zip` file from Barry for now.
b) Launch a terminal.
c) Use the `cd <directory-path>` command to navigate to the directory that contains `opto-ovj.zip`.
d) Execute the `unzip opto-ovj.zip` command to extract the `opto-ovj.zip` file.
e) Switch to the directory that was just extracted.
    a. If there is another version of OPTO present, run `sudo ./uninstall-opto.sh`
f) Install opto by running the `sudo ./install.sh <ovj-version>` command.
    a. For instance: `sudo ./install.sh`
g) If you wish, you can remove the files that were unzipped.``

3. Initial Setup:

Shimming: OPTO utilizes a text file named `opto-shim.dat` to define the shim names, their limits, and step size. A default file is included with the installation and can be found at `$vnmrsys/opto`. This file can be modified in the terminal through `gedit opto-shim.dat` command in the terminal command line.

Parameter Optimization: OPTO uses a parameter file located at `$vnmrsys/opto` folder. Default parameter files are provided with the installation for hC CP pulse program (NMRFAM use only for now, we can provide in the installation if needed). Initially, users should copy this file with a new name to the `$vnmrsys/opto` folder and then modify this file to meet their needs by following these steps:

a) Open the experiment that needs to be optimized.
b) List the parameters in the text files along with their minimum, maximum, and step size.

c) Include all the relevant parameters that are defined in your pulse program, their global limits, and a reasonable step size.

The `$vnmrsys/opto` folder should contain parameter files tailored to each of your pulse sequences of interest.

3.1 SELECTING THE REGION OF INTEREST FOR THE INTEGRATION:

When the "*Freq*" option is selected, OPTO utilizes the integrated region using the normal region integration tool. To define and save a region of interest:

a) Acquire a 1D spectrum.
b) Assign the region you wish to integrate by clicking integration button along sidebar.
c) To integrate specific regions, click on spectra to place a line and type 'z' in the command line interface or type 'z(10p) z (5p)' to only use a cut region between 10 and 5 ppm
d) Define the region of interest.

OPTO reads this value when integrating the signal region during operation. It may be changed at any time during operation,

Otherwise, clear the integrated region by typing 'cz' in the command line. This will integrate the full spectral width.

If multiple regions are stored, OPTO only assesses the first region.

4. COMMUNICATION BETWEEN OPTO AND OPENVNMRJ:

a) Within OpenVnmrJ command line, type 'Listenon' to allow OpenVnmrJ to communicate to OPTO.
b) In terminal, create a file with the name of Opto-Parameters.dat
    a. Examples files are provided within the download files of OPTO
c) The file should have the format shown in Box 1.
d) These will be the parameter names OPTO will find within the software that it will edit. The ranges and step sizes may be changed here for OPTO operation. There can be an arbitrary number of these files made that correspond to the experiments you wish to perform.

- To open OPTO, type opto.sh in the command line.
- A popup window will appear where you need to select the parameter.dat file.
- Once the parameter file is selected, the OPTO window will appear.

Box 1: OPTO-parameter-hX.dat file layout.

```
#name, nickname, min, max, step
aHhx,,0,4095,1
aXhx,,0,4095,1
dHX,,-1000,1000,1
bHX,,0,1000,1
tHX,,0,15000,1
```

5. RUNNING OPTO FOR SHIMMING:

Prerequisites: Have a valid `opto-shim` and a `parameter file.dat` file, standard sample loaded and MA adjusted, initial spectra is acquired.

a) Have inserted in the probe the shimming standard. Spin sample up to the desired spinning rate.
b) Zero out all shimming values. Select shimming tab from the top line and select desired shims to optimize (a working example is described in text with additional details; a shorter example is below for quick testing).
c) Load previously used ^{1}H to ^{13}C cross polarization and decoupling parameters for the current spinning rate into the spectrometer. Ensure the decoupling power and acquisition time are safe.
d) Acquire a single spectrum of adamantane without changing shims using current set of shim parameters for comparison to OPTO.
e) Set the acquisition time to 50 ms, or alternatively, on the OPTO GUI set the **Before** command to issue to the spectrometer software the acquisition time to be 50ms.
f) In pass 1 select the first order shims (x^1, y^1, z^1).
g) In pass 2 select the second order shims (xy, xz, yz, z^2, x^2y^2).
h) In pass 3 select the third order shims (x^3, y^3, z^3, xz^2, yz^2, zxy, zx^2y^2).
i) In pass 4 select the highest order of shims (z^4, z^3x, z^3y, z^5, z^4x, z^4y).
j) For additional improvements to the shimming, run the above passes 1-4 twice more after they have converged with setting the acquisition time to 200 ms for the first replicate, and then set the acquisition time to 400 ms for the last replicate.
k) Use default number of iterations, range, and tolerance for experiment. Click **Run Job**.
l) Confirm that the spectrometer is performing the experiment and is pulsing. The spectrometer controlling software should be updating as the experiments are running live.
m) Once the Run is complete: If operating on the OvnmrJ software, run the provided macro 'gosopto(*savefilename*)' to update the parameters and perform a final run of the experiment with the optimal values and save the resulting experiment to file.
n) If wanted, the .log files of the optimization are written out to the experiment relevant directory for *posthoc* processing and analysis.

6. RUNNING OPTO FOR CP:

a) **Sample Preparation:**
    a. Have inserted in the probe the Cross Polarization standard. Spin sample up to the desired spinning rate. We use crystalline N-acetyl-Valine for setting CP conditions (between 2 kHz to 24 kHz ω_r) or microcrystalline GB1(25kHz +)
b) **Experiment Setup:**
    a. Open OPTO with opto.sh command.
    b. Load pulse sequence with ^{1}H to ^{13}C cross polarization option. (We use the hX pulse sequence with its associated parameters for this example). Ensure the parameter

names and ranges within the Opto-Parameters.dat file reflect the names of the variables used in the pulse sequence. An example is shown in Box 1.

c. Check that decoupling parameters are safe for the acquisition time chosen.
d. Estimate the power levels wanted for the Hartmann-Hahn condition for the sample in the spectrometer. Depending on the spinning rate chosen, these can be double quantum or single quantum conditions.

c) **Initial Spectrum Acquisition:**
   a. Here the user is expected to have a theoretical starting point get an initial CP signal. A previously optimized CP experiment can be a good starting point here.
   b. Acquire a 1D cross polarization spectrum for comparison.
   c. Once the unoptimized 1D is finished and before running OPTO job, ensure that you are in the correct experimental page in the OPTO GUI. Press **Resync** otherwise and check parameters.

d) **Select Region of Interest:**
   a. Select the region of interest and using section 4.1 above.

e) **Edit Parameter File:**
   a. Open the `opto-parameters.dat` file located in `~/vnmrsys/opto.`
   b. Edit the file to enter the parameters you want to optimize. Save it under a different filename in the same folder (example: "`opto-parameters_CP.dat`"). Unused parameters can be removed or commented out.

      Note: Ensure that all parameter names used are the same naming convention as in the maclib files.

f) **Set Cross Polarization optimization job:**
   a. Select the 'Freq' integration method for best CP optimization results. Ensure the correct region of spectrum is selected.
   b. In pass 1, select the parameters that control the power levels of the ^{1}H and ^{13}C channels.
   c. In pass 2, select the parameters that control the power levels of the ^{1}H and ^{13}C channels and the ramp power delta.
   d. In pass 3, select the parameters that control the power levels of the ^{1}H and ^{13}C channels, the ramp power delta, and the ramp shape.
   e. Check that the spectrometer software has the appropriate number of scans, dummy scans, acquisition time, and integration range selected for operation. If not, these parameters may be written into the **Before** tab to be issued to the spectrometer before operation.
   f. Once the parameters and commands are set and the operator is happy with how they expect the Job to be performed, one may save the entirety of the job using the **Save Job** button and then can load the optimization protocol to a different experiment with the **Load Job** button.

g) **Run Optimization:**
   a. Use the default settings for the tolerance and range change them if needed.
   b. Click on '`Run job`'.
   c. The results written out to the terminal opto.sh was launched from as the run progresses.

h) **Completion and Application:**

a. Once the Run is complete: If operating on the OvnmrJ software, run the provided macro `gosopto(<savefilename>)` to update the parameters and perform a final run of the experiment with the optimal values and save the resulting experiment to file.

b. If wanted, the .log files of the optimization are written out to the experiment relevant directory for *posthoc* processing and analysis.